

AKHILA BODEPUDI

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PROFESSIONAL SUMMARY

Python Full Stack Developer with 3+ years of experience designing, building, and supporting enterprise-grade applications across backend and frontend development. Proficient in **Python**, **FastAPI**, **Django**, and **Flask** for backend services, with strong frontend experience in **React.js** and **Angular**. Skilled in **RESTful API** design, **microservices architecture**, and cloud-based deployment using **Docker**, **Kubernetes**, **AWS**, and **Azure**, with exposure to **CI/CD pipelines** using **GitHub Actions** and **Jenkins**. Holds a **Master of Science in Computer Science** from the **University of Maryland, Baltimore County**, along with **Azure AI Engineer** and **Google Cloud Associate** certifications. Known for a collaborative, detail-oriented approach to writing lean, testable, and scalable code within agile teams.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, SQL, HTML5, CSS3

Backend Frameworks: FastAPI, Django, Django REST Framework, Flask, Django ORM, Pydantic

Frontend Frameworks: React.js, Angular, Redux, Angular Material, Vite, Bootstrap, Leaflet.js

Databases: PostgreSQL, SQLite, MySQL, SQL Server, MongoDB, Redis

Cloud Platforms: AWS (EC2, S3, Lambda), Azure (App Service, Azure Functions, Azure Kubernetes Service), Vercel, Render

AI & Machine Learning: OpenAI API, Retrieval-Augmented Generation (RAG), XGBoost, scikit-learn, NumPy, Pandas, joblib

DevOps & CI/CD: GitHub Actions, Jenkins, CI/CD Pipelines, Docker Compose

Testing: pytest, Postman, Unit Testing, Integration Testing

Frontend Technologies: HTML5, CSS3, Bootstrap, Django Templates

Data Processing: Pandas, NumPy

API Development: RESTful APIs, Microservices Architecture, Input Validation, Exception Handling, CORS

Containerization & Orchestration: Docker, Docker Compose, Kubernetes, Nginx

Practices: Agile, Collaborative Development, Code Review, Cross-functional Collaboration

PROFESSIONAL EXPERIENCE

Python Full Stack Developer | Cisco Systems, USA

Jan 2026 - Present

- Contributed to the development of backend services for Cisco's enterprise network management platform using **Python**, **FastAPI**, and **Django REST Framework** within a **microservices architecture**, supporting a 20% improvement in backend processing efficiency.
- Assisted in developing **RESTful APIs** to integrate backend services with **Angular** applications and **SQL** databases, implementing input validation and exception handling under the guidance of senior engineers.
- Developed frontend components using **Angular**, **TypeScript**, and **Angular Material** to support network monitoring and configuration dashboards utilized by internal engineering teams.
- Supported data processing workflows using **Python**, **SQL**, **Pandas**, and **NumPy** to prepare operational network data for analytics and reporting.
- Assisted in containerizing application services using **Docker** and gained hands-on experience deploying **microservices** with **Kubernetes** on **AWS** across development and testing environments.
- Supported automated testing and **CI/CD** processes using **pytest**, **Postman**, **GitHub Actions**, and **Jenkins**, contributing to the release and deployment workflow.

Python Full Stack Developer | Hexaware Technologies, India

Jun 2022 - Aug 2024

- Contributed to enterprise full-stack applications using **Python**, **Django**, and **React.js**, supporting business applications used across customer and workflow management functions.
- Assisted in developing **REST APIs** and **microservices** using **FastAPI** and **Flask**, collaborating with senior engineers on design, implementation, and integration for 5+ backend services covering customer management and workflow automation.
- Helped build reusable frontend components using **Redux**, **HTML5**, **CSS3**, and **Bootstrap**, contributing to several application screens and integrating with backend APIs.
- Supported backend data-processing workflows using **Pandas** and **SQL**, assisting with cleaning and processing of enterprise datasets to improve reporting accuracy.
- Assisted in containerizing applications using **Docker**, and gained hands-on experience deploying services with **Kubernetes** on **Azure** across development and QA environments.
- Wrote unit and integration tests using **pytest** and **Postman** for assigned **REST endpoints**, contributing to overall test coverage across application releases.

ACADEMIC PROJECTS

AI-Powered Customer Support Assistant

Description: Developed a full-stack customer support assistant that answers questions from an approved knowledge base, shows source citations and confidence, and routes unsupported questions to a human instead of guessing.

Technologies: Python, FastAPI, React.js, TypeScript, RAG, OpenAI API, SQLite, Docker, Vercel, Render

- Developed backend APIs with Python and FastAPI for chat, document ingestion, feedback, and conversation history, using retrieval-augmented generation over company policy documents.
- Built a responsive React.js and TypeScript chat UI with citations, confidence labels, helpfulness feedback, and human-escalation actions.
- Implemented grounded retrieval (chunking, relevance ranking, and a minimum score threshold) so answers stay tied to verified sources and low-evidence questions escalate.
- Added an optional OpenAI generation path with context-only prompting, plus a zero-cost local extractive mode that runs without an API key.
- Containerized the app with Docker Compose and deployed the frontend on Vercel and the API on Render, with CI for pytest, Ruff, ESLint, and TypeScript builds.

AgroNity - Full-Stack Agriculture Marketplace & ML Platform

Description: Built and deployed a production Django web app that combines an online crop marketplace with a machine learning recommendation engine, giving farmers, customers, and admins one platform to buy, sell, and make data backed crop decisions.

Technologies: Python, Django, XGBoost, scikit-learn, NumPy, PostgreSQL, SQLite, HTML5, CSS3, Bootstrap 5, Leaflet.js, Vercel

- Designed and built the backend in Django, including custom user models, role-based auth (Farmer/Customer/Admin), and an admin approval workflow.
- Built the marketplace end to end (crop listing, editing, cart, and checkout) with ownership checks so users can only edit their own listings.
- Trained an XGBoost model on soil and weather data (N, P, K, temperature, humidity, pH, rainfall) that hit 99.5% accuracy for crop recommendations and wired it into the Django app for live predictions.
- Added a Leaflet/OpenStreetMap integration so farmers can locate nearby fertilizer shops and soil-testing centers.
- Handled deployment myself: set up Vercel for serverless hosting, migrated from SQLite to Neon PostgreSQL for production, and configured WhiteNoise for static file delivery.

EDUCATION:

Master of Science (M.S.) in Computer Science | May 2026 | University of Maryland, Baltimore County (UMBC) | USA

CERTIFICATIONS:

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| • Azure AI Engineer Associate Microsoft | • Google Cloud Associate Cloud Engineer |
| • Data Visualization TATA Group | • Cloud Computing NPTEL |